SURF CLOCK — COMPLETE PRUSASLICER SETTINGS MANUAL (PAGE 1 OF 2)

Every Exact PrusaSlicer Setting for Heat, Ironing, Walls, Seams & Combining Prints

1. CAN YOU COMBINE PARTS ON ONE PRINT BED?

- **YES! Hands + Snap Pins Combined (1 Click):**
  Use the new file 5_AND_6_COMBINED_Hands_Plus_Snap_Pins_PrusaMINI.stl in your folder! It combines BOTH Pointer Hands (#5) and ALL 12 Snap-Lock Pins (#6) onto one 77 x 51 mm plate so they print together in a single 32-minute job in Matte Orange PLA on your Prusa MINI (or MK4).
- **Why the 4 Large Plates Print Individually:**
  Files #1 (Outer Frame, 230x184mm), #2 (Dial Face, 222x176mm), #3 (Motor Mount, 222x176mm), and #4 (Back Cover, 222x176mm) each fill ~88% of the Prusa MK4 bed (250x210mm), so those 4 print one per bed.

2. DO YOU NEED TO CHANGE HEAT / TEMPERATURES?

- **Select Profile: Generic PLA (or Prusament PLA):**
  PrusaSlicer's built-in Generic PLA profile automatically sets Nozzle = 215°C (First Layer) / 210°C (Other Layers) and Heated Bed = 60°C. You do NOT need to manually override temperatures for White, Black, or Orange PLA.
- **ONE Temperature Tip for Overture Matte White / Matte Orange:**
  Matte PLA looks dead-flat and velvety (zero plastic shine) when printed at 205°C - 210°C. Stick with Generic PLA (215°C first layer / 210°C other layers, 60°C bed, 100% Cooling Fan).
- **ONE Temperature Tip if Using Transparent Clear PLA on File #4:**
  For crystal-clear light transmission on the Back Cover (#4), raise Other Layers Nozzle Temp to 220°C and reduce Cooling Fan to 30% so layers fuse like glass.

3. WHAT ABOUT IRONING? (CRITICAL FOR MUSEUM FINISH!)

- **Where to Find Ironing in PrusaSlicer:**
  Click the [Print Settings] tab at the top -> Click [Infill] on the left menu -> Scroll down to the [Ironing] section.
- **TURN IRONING ON for File #2 (Dial Face) & File #5 (Pointer Hands):**
  Check [x] Enable ironing. Set Ironing type = [All top surfaces], Flow rate = [15%], Spacing between ironing passes = [0.10 mm], Ironing speed = [15 mm/s].
- **Why [All top surfaces] is Magic on File #2 (Dial Face):**
  Your Dial Face has a flat White background at Z = 2.40 mm and raised Black letters at Z = 3.00 mm. With [All top surfaces] enabled, the nozzle irons the entire White dial face to a silky porcelain finish right at Z = 2.40 mm BEFORE pausing at Z = 2.60 mm for your Black filament swap, and then irons the tops of the Black letters at Z = 3.00 mm!
- **TURN IRONING OFF for File #1 (Outer Frame) & File #3 (Motor Mount):**
  Leave Ironing UNCHECKED on File #1 (Outer Frame) and File #3 (Motor Mount). Why? Because File #1 prints FACE-DOWN on the PEI sheet! The visible front bezel of the frame is at Z = 0.00 mm against the Prusa Satin/Textured PEI bed, which already gives it a factory-molded matte texture without wasting 40 minutes ironing the hidden rear rim.

4. EXACT PRUSASLICER CHECKLIST TO MAKE IT LOOK AMAZING

- **A. Perimeter Generator = Arachne (Print Settings -> Layers and perimeters):**
  Ensure Perimeter generator is set to [Arachne] (this is default in PrusaSlicer 2.6+). Arachne dynamically adjusts extrusion width so the 1.04 mm Braun DIN lettering and chapter ring ticks print with zero gaps or blobs.
- **B. Seam Position = Rear (Print Settings -> Layers and perimeters):**
  Set Seam position = [Rear] (or [Aligned]). This hides the tiny Z-seam line along the top/rear corner edge instead of scattering random dots across the side walls of your white frame.
- **C. Top & Bottom Fill Pattern = Monotonic Lines (Print Settings -> Infill):**
  Ensure Top fill pattern = [Monotonic lines] and Bottom fill pattern = [Monotonic]. This guarantees every stroke is laid down left-to-right in the exact same direction so there are never any light/dark striping bands across the dial face.
- **D. Perimeters = 3 & Solid Layers = 5 Top / 4 Bottom:**
  In [Layers and perimeters], set Perimeters = [3], Top solid layers = [5], Bottom solid layers = [4]. Having 5 top solid layers prevents the 15% infill grid from ever ghosting (showing through) under white PLA!

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Part-by-Part Cheat Sheet & The 3-Click Z = 2.60 mm Color Swap Guide

5. EXACT SETTINGS FOR EACH OF YOUR 5 PRINT JOBS

PRINT 1 — 1_Outer_Frame_MATTE_WHITE_PrusaMK4.stl (Prusa MK4, ~3h 10m):
- Spool: Overture Matte White PLA | Preset: 0.20mm STRUCTURAL | Nozzle: 215°C/210°C | Bed: 60°C
- Perimeters: 3 | Infill: 15% Gyroid | Supports: NONE | Brim: OFF | Ironing: OFF (Front bezel is face-down on bed at Z=0.00mm!) | Seam position: Aligned or Rear.

PRINT 2 — 2_Dial_Face_WHITE_then_BLACK_at_Z2.6mm_PrusaMK4.stl (Prusa MK4, ~55m):
- Spool: Start with Matte White PLA -> Swap to Matte Black PLA at Z = 2.60 mm (Layer 13)
- Preset: 0.20mm STRUCTURAL (or 0.20mm QUALITY) | Perimeters: 3 | Top Solid Layers: 5 | Infill: 20% Grid
- Ironing: ON ([x] Enable ironing -> Type: All top surfaces, Flow: 15%, Spacing: 0.10mm) | Arachne: ON.

PRINT 3 — 3_Motor_Mount_Plate_ORANGE_or_WHITE_PrusaMK4.stl (Prusa MK4, ~45m):
- Spool: Matte Orange (or White) PLA | Preset: 0.20mm STRUCTURAL | Perimeters: 3 | Infill: 20% Grid | Supports: NONE | Ironing: OFF (internal functional chassis plate).

PRINT 4 — 4_Back_Cover_TRANSPARENT_or_WHITE_PrusaMK4.stl (Prusa MK4, ~30m):
- If printing in Transparent Clear PLA: Preset 0.20mm QUALITY, Perimeters: 3, Top/Bottom Solid Layers: 2, Infill: 20% Aligned Rectilinear, Nozzle Temp: 220°C, Fan: 30%, Ironing: ON (Topmost surface only).
- If printing in Solid White PLA: Same settings as Print 3 (0.20mm STRUCTURAL, 15% Infill, Ironing OFF).

PRINT 5 — 5_AND_6_COMBINED_Hands_Plus_Snap_Pins_PrusaMINI.stl (Prusa MINI or MK4, ~32m):
- Spool: Matte Orange PLA | Preset: 0.15mm QUALITY or 0.20mm QUALITY | Perimeters: 3 | Infill: 25%
- Ironing: ON ([x] Enable ironing -> All top surfaces, Flow: 15%) so the orange pointer blades look like painted metal!
- IMPORTANT: Let the steel PEI sheet cool down for 2 full minutes before removing the hands so the thin needles never bend while warm.

6. HOW TO DO THE 2-COLOR DIAL SWAP AT Z = 2.60 mm (STEP-BY-STEP)

Step 1 — Drag & Slice:
Drag 2_Dial_Face_WHITE_then_BLACK_at_Z2.6mm_PrusaMK4.stl onto the Prusa MK4 platter. Turn ON Ironing (All top surfaces) and set Top Solid Layers = 5. Click [Slice now] in the bottom right corner.

Step 2 — Drag the Right-Side Orange Layer Slider to 2.60 mm (Layer 13):
Look at the vertical orange ruler bar on the right side of the 3D view. Drag the top handle down until you see [2.60] (Layer 13). Notice how at 2.40 mm you see the flat white dial plate, and at 2.60 mm ONLY the raised letters and chapter ring appear!

Step 3 — Click the Orange [+] Sign:
Click the little orange [+] button directly to the left of [2.60]. A color-change marker appears at 2.60 mm. Click [Slice now] one final time and click [Export G-code].

Step 4 — At the Printer:
Load MATTE WHITE PLA to start the print. At Z = 2.40 mm, the MK4 will iron the entire white dial face smooth. Then at Z = 2.60 mm, the MK4 will beep, move the print head to the front corner, automatically spit out the White PLA, and ask you on the LCD screen to insert MATTE BLACK PLA. Push in the Black PLA, click Continue, and in 8 minutes your 2-color Braun dial is finished!